

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Computer Networks

COURSE CODE: CSA0703

COURSE OUTCOME COVERED

CO5: *Measure network performance using key metrics with simulation tools*

Assessment Tool Weightage: 20%

QUESTIONS: 40

Total Marks:40

Annexure A

Mock Interview

QUESTIONS:40

Total Marks:40

1. Analyze how network congestion affects packet delay and packet loss. (BL4)(1)
2. Compare the impact of star and mesh topologies on network reliability and fault tolerance. (BL4)(1)
3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically. (BL4)(1)
4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic. (BL4)(1)
5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication. (BL5)(1)
6. Analyze how network segmentation using VLANs improves performance and security. (BL4)(1)
7. Compare the effects of TCP and UDP on network performance for different applications. (BL4)(1)
8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability. (BL5)(1)
9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization. (BL4)(1)
10. Evaluate the trade-off between network security measures (such as firewalls and encryption) and overall network performance. (BL5)(1)
11. A network has high delay—how will you identify and fix the issue? (BL4)(1)
12. If throughput is low despite high bandwidth, what could be the reasons? (BL4)(1)
13. How would you reduce packet loss in a congested network? (BL4)(1)
14. A VoIP call is breaking frequently—how would you troubleshoot? (BL4)(1)
15. Given a network with high jitter, suggest improvements (BL4)(1)
16. How will you simulate a congested network in Packet Tracer? (BL5)(1)
17. If packets are delayed in a router, what steps will you take? (BL4)(1)
18. How would you optimize a network for online gaming? (BL4)(1)
19. Design a small network and explain how you will measure its performance (BL5)(1)
20. If RTT is very high, how will you diagnose the issue? (BL4)(1)
21. Explain network performance to a non-technical person (BL5)(1)
22. Describe throughput using a real-life example (BL5)(1)
23. Present how packet loss affects video streaming (BL5)(1)
24. Explain delay types clearly in under 1 minute (BL4)(1)
25. Describe Cisco Packet Tracer in a professional way (BL5)(1)
26. Explain QoS in simple and technical terms (BL5)(1)

- 27. How would you present your simulation results to a client? (BL4)(1)
- 28. Explain jitter using a real-world analogy (BL5)(1)
- 29. Describe how you tested network performance in a lab (BL4)(1)
- 30. Explain the importance of performance metrics in one structured answer (BL5)(1)
- 31. Introduce yourself and explain your knowledge of network performance (BL5)(1)
- 32. Why are you interested in networking and simulation tools? (BL5)(1)
- 33. What makes you confident in analyzing network performance? (BL5)(1)
- 34. Explain a concept you are very strong in (related to networking) (BL5)(1)
- 35. How would you handle a question you don't know? (BL5)(1)
- 36. Describe a situation where you solved a technical problem (BL4)(1)
- 37. Why should we select you for a networking role? (BL5)(1)
- 38. What challenges did you face while learning Packet Tracer? (BL5)(1)
- 39. How do you improve your technical skills regularly? (BL5)(1)
- 40. Demonstrate confidence while explaining a complex concept simply (BL5)(1)

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

Answers:

1. How does network congestion affect packet delay and packet loss?

Answer:

Network congestion occurs when too much traffic passes through the network. As a result, routers take longer to process packets, increasing delay. If router buffers become full, packets are dropped, causing packet loss. Congestion reduces throughput and affects applications like video calls and online gaming.

2. Compare star and mesh topologies on reliability and fault tolerance.

Answer:

A **star topology** is easy to manage and requires less cabling. If one device fails, the others continue working, but if the central switch fails, the entire network stops. A **mesh topology** provides multiple communication paths, making it highly reliable and fault tolerant, but it is more expensive and complex.

3. Why does increasing the number of routers increase end-to-end delay?

Answer:

Each router receives, processes, and forwards packets. Every router adds processing, queuing, and transmission delay. Therefore, more routers between the source and destination increase the total end-to-end delay.

4. How does QoS improve voice and video performance?

Answer:

Quality of Service (QoS) prioritizes important traffic like voice and video over normal data traffic. It reduces delay, jitter, and packet loss, ensuring smooth video conferencing and clear voice communication.

5. Evaluate fiber optic cables instead of copper cables.

Answer:

Fiber optic cables provide higher bandwidth, faster data transmission, longer distance support, and immunity to electromagnetic interference. Copper cables are cheaper but slower and suitable only for shorter distances.

6. How do VLANs improve performance and security?

Answer:

VLANs divide one physical network into multiple logical networks. This reduces unnecessary broadcast traffic, improves performance, and isolates users, increasing network security.

7. Compare TCP and UDP.

Answer:

TCP is reliable because it ensures packet delivery, error checking, and retransmission. It is suitable for web browsing and file transfer. UDP is faster because it does not guarantee delivery, making it ideal for video streaming, VoIP, and online gaming.

8. Why is load balancing important?

Answer:

Load balancing distributes network traffic across multiple servers or links. It prevents overload, improves performance, increases availability, and provides better fault tolerance.

9. How do frequent routing updates affect the network?

Answer:

Frequent routing updates help routers maintain accurate routing information but consume bandwidth and processing power. Too many updates may reduce network efficiency.

10. Security vs Network Performance

Answer:

Security measures like firewalls and encryption protect data but require additional processing, which may slightly increase delay. The goal is to balance strong security with acceptable performance.

11. A network has high delay. How will you identify and fix it?

Answer:

I would check bandwidth utilization, router CPU usage, cable quality, and routing paths using ping and traceroute. I would remove congestion, upgrade bandwidth if required, optimize routing, and apply QoS.

12. Why is throughput low despite high bandwidth?

Answer:

Possible reasons include congestion, packet loss, high latency, poor routing, hardware limitations, or protocol overhead. High bandwidth alone does not guarantee high throughput.

13. How would you reduce packet loss?

Answer:

Reduce congestion, increase bandwidth, replace faulty cables, optimize routing, configure QoS, and upgrade overloaded routers or switches.

14. A VoIP call is breaking. What will you do?

Answer:

I would check delay, jitter, packet loss, bandwidth usage, QoS settings, and network congestion. Prioritizing voice traffic usually improves call quality.

15. High jitter. What improvements will you suggest?

Answer:

Enable QoS, reduce congestion, increase bandwidth, avoid overloaded routers, and use stable wired connections wherever possible.

16. How will you simulate congestion in Packet Tracer?

Answer:

I would create multiple PCs sending large amounts of traffic through the same router or switch simultaneously, then observe delay, throughput, and packet loss.

17. Packets are delayed in a router. What steps will you take?

Answer:

Check CPU utilization, interface errors, routing table, queue size, bandwidth usage, and QoS configuration. Remove bottlenecks or upgrade the device if necessary.

18. How would you optimize a network for online gaming?

Answer:

Use QoS to prioritize gaming traffic, reduce latency, use wired Ethernet, minimize packet loss, and avoid unnecessary background traffic.

19. Design a small network and measure performance.

Answer:

I would connect two PCs through a switch and a router in Cisco Packet Tracer. I would use ping to measure delay, calculate throughput, monitor packet loss, and observe network utilization.

20. RTT is very high. How will you diagnose it?

Answer:

Use ping and traceroute to locate the delay, check congestion, inspect routing paths, verify bandwidth usage, and examine hardware performance.

21. Explain network performance to a non-technical person.

Answer:

Network performance is like road traffic. If the roads are clear, vehicles reach quickly. If roads are crowded, travel becomes slow. Similarly, a good network transfers data quickly and reliably.

22. Throughput using a real-life example.

Answer:

Imagine a water pipe. The pipe size represents bandwidth, while the amount of water flowing every second represents throughput. Even with a large pipe, blockages reduce water flow.

23. Packet loss affects video streaming.

Answer:

Packet loss causes missing data during transmission, leading to frozen video, buffering, poor image quality, and interrupted playback.

24. Explain delay types in under one minute.

Answer:

There are four main delays:

- Processing delay – time to examine packets.
- Queuing delay – waiting in router queues.
- Transmission delay – sending bits onto the link.
- Propagation delay – travel time through the medium.

25. Describe Cisco Packet Tracer professionally.

Answer:

Cisco Packet Tracer is a network simulation tool developed by Cisco. It allows users to design, configure, simulate, and troubleshoot computer networks without requiring physical networking equipment.

26. Explain QoS simply and technically.

Answer:

Simple: QoS gives priority to important network traffic.

Technical: QoS classifies, marks, queues, and schedules packets to reduce delay, jitter, and packet loss for critical applications.

27. How would you present simulation results to a client?

Answer:

I would use graphs and tables to compare throughput, delay, packet loss, and bandwidth utilization before and after optimization, then explain the improvements in simple language.

28. Explain jitter using an analogy.

Answer:

Imagine buses arriving every five minutes. If they arrive at irregular intervals, passengers wait unpredictably. That variation is similar to network jitter.

29. How did you test network performance in a lab?

Answer:

I designed a network in Cisco Packet Tracer, generated traffic, measured throughput, delay, packet loss, and RTT using simulation mode and ping, then analysed the results.

30. Importance of performance metrics.

Answer:

Performance metrics like throughput, delay, packet loss, jitter, and RTT help evaluate network efficiency, identify problems, improve user experience, and support better network planning.

31. Introduce yourself.

Answer:

"My name is Sindhu. I am pursuing a B.Tech in Artificial Intelligence and Data Science. I have knowledge of networking concepts, Cisco Packet Tracer, and network performance analysis. I enjoy learning about network design, troubleshooting, and simulation, and I am eager to apply my skills in real-world projects."

32. Why are you interested in networking?

Answer:

I enjoy understanding how devices communicate and how networks support modern applications. Simulation tools like Cisco Packet Tracer help me learn practical networking without requiring expensive hardware.

33. What makes you confident in analysing network performance?

Answer:

I understand key metrics like throughput, delay, jitter, RTT, and packet loss. I have practised analysing these metrics using Cisco Packet Tracer and know how they affect network performance.

34. Explain a networking concept you are strong in.

Answer:

I am confident in explaining **Quality of Service (QoS)** because I understand how it prioritizes important traffic, reduces delay and jitter, and improves voice and video communication.

35. How would you handle a question you don't know?

Answer:

I would honestly admit that I am not completely sure, explain what I do know, and mention that I would quickly research the topic and learn the correct answer.

36. Describe a technical problem you solved.

Answer:

While using Cisco Packet Tracer, I faced a connectivity issue because of incorrect IP addressing. I verified the IP configuration, corrected the subnet mask and default gateway, tested using ping, and successfully restored connectivity.

37. Why should we select you?

Answer:

I have a strong willingness to learn, good problem-solving skills, and practical knowledge of networking concepts and Cisco Packet Tracer. I work well in teams and adapt quickly to new technologies.

38. Challenges while learning Packet Tracer.

Answer:

Initially, I found IP addressing, routing, and device configuration challenging. Through regular practice and troubleshooting different network topologies, I became more confident.

39. How do you improve your technical skills?

Answer:

I regularly practise in Cisco Packet Tracer, complete online courses, read networking documentation, watch technical tutorials, and work on small networking projects.

40. Demonstrate confidence while explaining a complex concept simply.

Answer:

"Think of the internet like a postal service. Data is divided into small packets, each carrying an address. Routers act like post offices, forwarding packets along the best route. Even if packets take different paths, they are reassembled correctly at the destination. This process allows fast and reliable communication across the world."